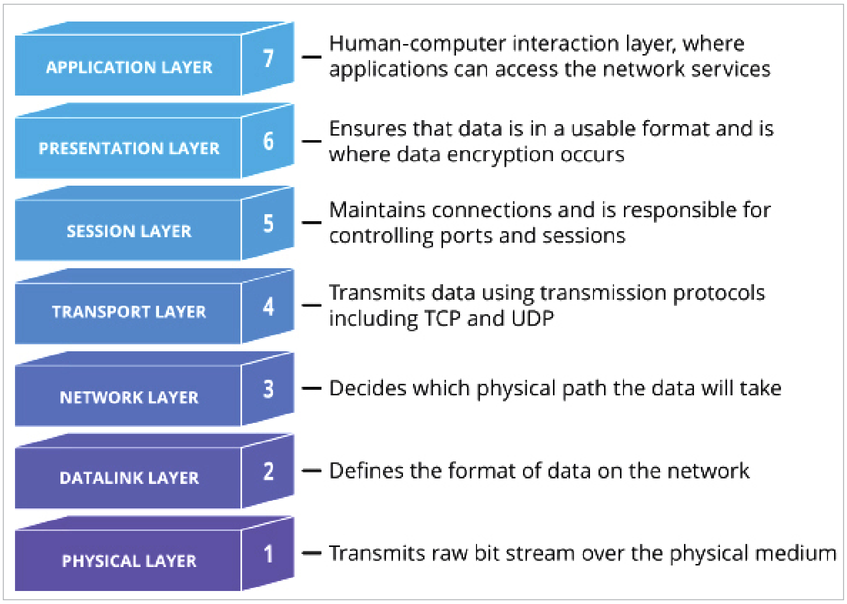

Figure 1: Open system interconnection model (Source: *https://www.cloudflare.com/en-in/learning/ddos/glossary/open-systems-interconnection-model-osi/*)

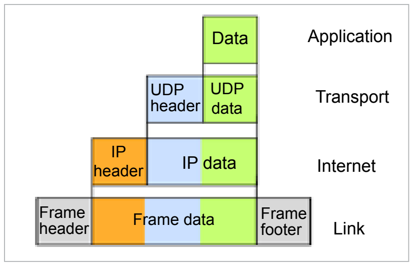

Figure 2: Data segmentation at each level (Source: *https://www.javatpoint.com/ip*)

- In the **presentation layer**, the messaging gets done. This layer makes sure the data goes through the required transition to be passed to the next layer such as compression/decompression, encryption/decryption, and translation from ASCII to EBCDIC.
- The **session layer** is responsible for the communication channel. If we have 1GB data to transfer, it takes control and makes sure the channel is open until the whole message is transferred.
- If we are using HTTP v2, which is on top of the TCP protocol, the **transport layer** helps to get the actual message split into chunks. It controls the flow and manages errors. Flow is controlled by sending the traffic according to the receiver bandwidth. If the segments/message chunks are transmitted to the sender, the layer tries to send them again, thus managing errors. This layer is said to be the heart of the model. Above it are all the software layers and below it are hardware layers.
- When we have a message to be transported across a network, we need the **network layer.** Since *google.com* is not in our network, in this layer the segments are shortened into packets and find the best path to the destination network. This is commonly known as routing.
- In the **data link layer**, the data is broken down further from packets to frames. The main function of this layer is to make sure data transfer is error-free from one node to another. When a packet arrives in a network, this layer is responsible for transmitting it to the host using its MAC address.
- **The physical layer** is responsible for the actual physical connection between the devices. Here the data gets converted into a bit stream, which is a string of 1s and 0s.

Just the reverse takes place when the message hits the Google server. Here, the received signal is converted into a bit stream, from frame to packet to segment to conversion and, finally, the message is sent to the application layer. Figure 2 gives a glimpse of how the data is segmented at each level and adds a header to it.

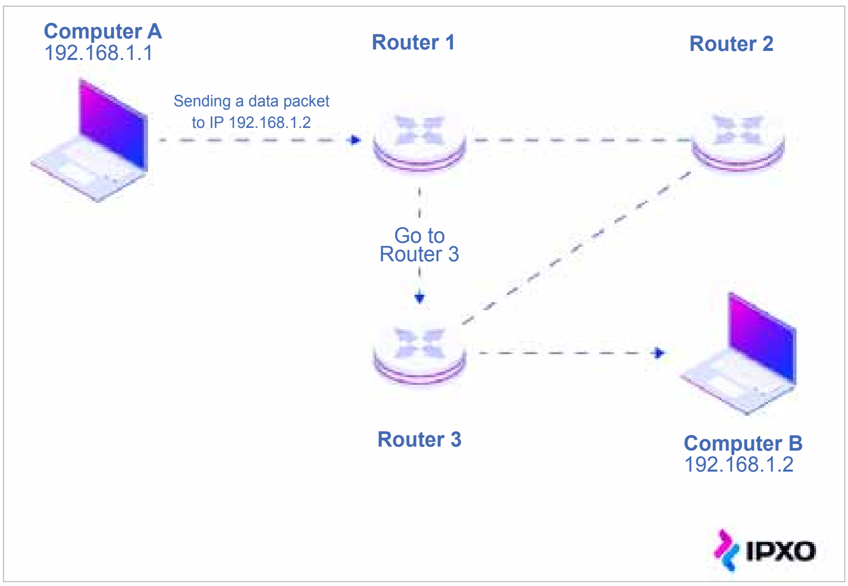

Figure 3: Network layer (Source: *https://www.ipxo.com/blog/network-routing/*)